Shivam Patil

(+91) 7038573273 | shivampatilinfo@gmail.com | Mumbai, Maharashtra | linkedin.com/in/shivam--patil | github.com/Git-ShivamPatil | shivamsfolio.com

Summary

Backend and data engineer at TCS with 1.5+ years building REST APIs in Python and FastAPI and distributed data processing with PySpark, and debugging production issues. Amazon Web Services Certified Developer - Associate.

Technical Skills

- **Languages:** Python, SQL, Java, TypeScript
- **Backend and APIs:** FastAPI, REST APIs, JWT, OAuth2, LDAP, Oracle Database 19c, Postman
- **Data and Distributed Processing:** PySpark, Spark SQL, Apache Hadoop (HDFS), Hive, Impala, Cloudera CDP
- **ML and AI:** PyTorch, OpenCV, YOLOv11, TensorFlow, Keras, Scikit-learn
- **Tools:** GitHub Actions, GitLab CI, pytest, Git, Cloudera Manager, Databricks, Delta Lake, AWS (certified)

Experience

Tata Consultancy Services (TCS) **Jan 2025 – Present**
Assistant Systems Engineer Mumbai, Maharashtra
Client: RBI CIMS

- Shipped **13+ load-balanced REST APIs** in **Python and FastAPI** for production CAS, DBIE portals serving **5,000+ requests/day**, cutting **p95 latency from 420 ms to 375 ms** via payload optimization.
- Automated **XML and spreadsheet parsing** plus **quality and outlier checks** in **PySpark** with partitioning and caching on **Cloudera CDP** for **11+ new returns**, cutting **manual checks by 13%**.
- Implemented element-level and cross-element **validation rules in SQL** on **Oracle Database 19c** with the team, catching reconciliation breaks and referential errors and cutting **submission failures from 28% to 19%**.
- Debugged **20-25 production issues** (3-4 large) across the FastAPI APIs, PySpark jobs and SAP BusinessObjects reports, tracing **root causes through logs and data** while monitoring jobs in **Cloudera Manager**.
- Wrote **pytest** tests for the PySpark jobs and built a new **GitHub Actions** CI pipeline running them on every push, updating existing **GitLab CI** and GitHub Actions pipelines.
- Built **JWT, OAuth2 and LDAP authentication** with **role-based authorization** for the CAS, DBIE portals per organization security policy, and fixed **security audit findings**.

Projects

BFSI Lending Lakehouse **Aug 2026**
Personal project | PySpark, Databricks, Delta Lake, Unity Catalog, Spark SQL

- Computed credit metrics in **Spark SQL** on **Databricks** (DPD buckets, roll-rate matrices, vintage curves, collection efficiency, PAR-30/60/90, Ind AS 109 ECL staging) over a **150,000-loan synthetic** no-cost-EMI book.

Autonomous Driving System **Sep 2026**
Personal project | PyTorch, OpenCV, YOLOv11

- Built a self-driving car simulation with **PyTorch and OpenCV**, using a custom **CNN** trained on **70,000+ camera images** to predict steering angles and interpret road environments.
- Developed custom evaluation metrics and used simulation tools to validate and enhance model performance across **1,000+ test scenarios** and multiple datasets.

Education

Zeal College of Engineering, Pune **Jun 2020 – 2024**
Bachelor of Technology in Artificial Intelligence and Data Science Pune, Maharashtra
CGPA: 8.8/10 | Coursework: Data Structures and Algorithms, Database Management Systems
LeetCode: 75+ problems across 13+ patterns | leetcode.com/u/shivam2op

Certifications

Amazon Web Services Certified Developer - Associate Link **Aug 2026**
Anthropic Claude Certified Architect - Professional Link **Sep 2026**

Leadership

Department President, Zeal Institutes **Aug 2023 – Jun 2024**